Physics 7A Discussion 3

Fall 2025   Sept 10–11

Topic Overview

- Motion in two dimensions by splitting into independent x and y components
- Projectile motion as a special case: constant horizontal velocity and constant vertical acceleration

Key Formulas

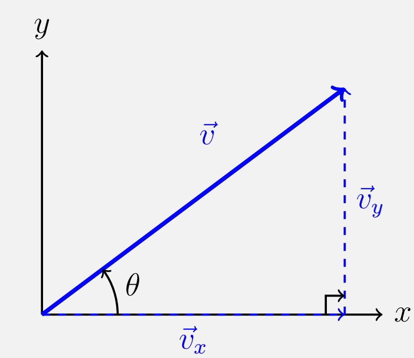
- Constant acceleration kinematics (applies in x or y direction separately):

$$\begin{aligned} v_f &= v_0 + at \\ \Delta d &= v_0 t + \tfrac{1}{2}at^2 \\ \Delta d &= v_f t - \tfrac{1}{2}at^2 \\ v_f^2 &= v_0^2 + 2a\Delta d \\ \Delta d &= \tfrac{1}{2}(v_0 + v_f)t \end{aligned}$$

- Velocity components:

$$v_{0x} = v_0 \cos\theta, \qquad v_{0y} = v_0 \sin\theta$$

- Vector decomposition of initial velocity:



Part 1: Conceptual Questions

a) Why is 2D motion considered a superposition of independent horizontal and vertical motions?

b) Suppose two balls are released simultaneously (assume no air resistance): one is dropped straight down, the other is launched horizontally from the same height. Which hits the ground first? Explain.

c) Can a projectile's horizontal velocity ever change (ignoring air resistance)? Why or why not?

Part 2: Basic Problems

a) A ball is launched horizontally at $10\,\mathrm{m/s}$ from a $20\,\mathrm{m}$ tall cliff.

(i) How long does it take to hit the ground?

(ii) How far from the base does it land?

b) A soccer player kicks a ball at $v_0 = 15\,\mathrm{m/s}$ at an angle of 30° above the horizontal.

(i) Write expressions for $x(t)$ and $y(t)$.

(ii) At what time does the ball reach its maximum height?

(iii) What is that maximum height?

Part 3: Word Problems

a) A firefighter aims a hose at a burning window 10 m above the ground and 8 m away horizontally. If the water exits the hose at 20 m/s, at what angle(s) should the hose be aimed? (Neglect air resistance.)

b) A toy car is launched from the origin with initial velocity components $v_{0x} = 2.0\,\mathrm{m/s}$ and $v_{0y} = 4.0\,\mathrm{m/s}$. It experiences a constant horizontal acceleration $a_x = 1.0\,\mathrm{m/s^2}$ (as if a steady wind is pushing it), while the vertical motion is under gravity with $g = 9.8\,\mathrm{m/s^2}$.

(i) Write the equations for $x(t)$ and $y(t)$.

(ii) How long does it take for the car to return to the ground ($y = 0$)?

(iii) What is its horizontal displacement at that time?

c) A boat wishes to cross a river 200 m wide (that is, from the south bank to the north bank) that flows east at 3.0 m/s. The boat can move at 5.0 m/s relative to the water.

(i) If the boat heads straight north (perpendicular to the banks), it will still drift east because the river flows. How far downstream does it land on the opposite bank?

(ii) To arrive directly across from its starting point, the boat must aim at an angle west of north so that its velocity relative to the water cancels the river's flow. What angle (relative to north) must the boat aim?

(iii) Compare the time to cross in the two cases. (Hint: the crossing time depends only on the northward component of the boat's velocity.)

Part 4: Formula Derivations

a) Derive the formula for the range R of a projectile launched from the ground with initial speed v_0 and angle θ. What launch angle maximizes the range?

b) A projectile is launched (at an angle θ) up a hill that makes an angle ϕ with the horizontal. Derive an expression for the range along the slope in terms of v_0, θ, ϕ, and g.

c) A ball must clear a wall of height H located a distance D away. Find the condition on v_0 and θ such that the ball just passes over the wall.

d) For a fixed launch speed v_0, show that there are generally two launch angles that hit the same target at distance d. What is the relationship between these two angles?